TFPT Note S1

Solutions to Foundational Problems in Physics

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Abstract

This note outlines five solutions to open problems in fundamental physics derived from the Topological Fixed Point Theory (TFPT) framework. We demonstrate how Asymptotic Safety provides a UV completion for gravity with torsion, how spacetime singularities are resolved via a torsion-driven bounce, and how black holes can be understood as non-singular spacetime vortices. Furthermore, we present the TFPT axion as a dark matter candidate with a fixed misalignment angle and derive dark energy from either a torsional condensate or the terminal stage of the E8 cascade. All results are derived from the core TFPT invariants (c_3, ϕ_0) without introducing new free parameters, yielding sharp, testable predictions.

1 UV Completion via Asymptotic Safety

Ansatz (UFE-Compatible). We use the effective average action (FRG) with an independent torsion field, truncated to operators present in the Unified Field Equation (UFE):

$$\Gamma_k = \int d^4x\sqrt{-g}\left[Z_R(k)\,R - Z_{\nabla K}(k)\,\nabla\cdot K + Z_{K^2}(k)\,K^2 - 2\Lambda_k\right] + \mathcal{L}_{a,\mathrm{EM}}, \tag{1}$$

where $\mathcal{L}_{a,\mathrm{EM}} = -\frac{1}{4}F^2 + \frac{1}{2}(\partial a)^2 + c_3\,a\,F\tilde{F}$. The axion-photon coupling $g_{a\gamma\gamma} = -4c_3$ is fixed by the cubic condition, with $c_3 = 1/(8\pi)$ being a core TFPT invariant. The UFE itself is

$$\left(R - \nabla\cdot K + K^2\right)_{AB} = \hat{\kappa}^2\left(T^{(a)}_{AB} + T^{(\mathrm{EM})}_{AB}\right). \tag{2}$$

Dimensionless Couplings. The dimensionless couplings are defined as $g(k) = k^2/(8\pi\,Z_R(k))$, $\lambda(k) = \Lambda_k/k^2$, $\tau(k) = Z_{K^2}(k)/Z_R(k)$, and $\chi(k) = Z_{\nabla K}(k)/Z_R(k)$. The axion-photon coupling $y \equiv g_{a\gamma\gamma} = -4c_3$ is scale-invariant.

Wetterich Equation and Flows (truncated to $\{g, \lambda, \tau\}$). The flow equations take the form $\partial_t g = (2+\eta_R)g$, $\partial_t\lambda = (-2+\eta_R)\lambda + c_g\,g + c_\tau\,g\,\tau$, and $\partial_t\tau = A_\tau g - B_\tau\tau + \mathcal{O}(g\tau, \tau^2)$, with the anomalous dimension $\eta_R = a_1 g + a_2 g\tau + a_3 g\,y^2 + \ldots$. Crucially:

- The K^2 term is positive definite, ensuring $A_\tau > 0$ and $B_\tau > 0$, which means torsion provides UV damping.
- The axion contribution via $y^2 = 16c_3^2 \approx 0.0253$ is small and fixed, preserving the flow structure.

Existence Theorem (Non-Trivial UV Fixed Point). Since $\partial_t g \approx 2g > 0$ for $g \to 0$ and $\eta_R \sim -|a_1|g$ for large g, the intermediate value theorem guarantees a fixed point $g_\star > 0$ where $2 + \eta_R(g_\star, \tau_\star) = 0$. The other fixed-point equations then yield finite values $\tau_\star > 0$ and $\lambda_\star$. This establishes a non-Gaussian UV fixed point $(g_\star, \lambda_\star, \tau_\star)$ whose coordinates depend only on the TFPT invariants (c_3, ϕ_0) and field content, leaving no free parameters.

Testable Consequence. The same structure that supports the UV fixed point fixes the cosmic birefringence to be

$$\beta = \frac{\phi_0}{4\pi} = 0.2427^\circ. \tag{3}$$

This prediction is consistent with Planck PR4 data and provides a direct observational test of the theory.

2 Singularity Resolution via Torsion and Bounce

Modified Friedmann Equation. For a homogeneous and isotropic (FLRW) geometry with a time-dependent axial torsion field $S(t)$, the 00-component of the UFE yields a modified Friedmann equation:

$$H^2 = \frac{\kappa^2}{3}\,\rho - \frac{1}{3}\,\mathcal{K}^2 + \frac{\Lambda_{\text{eff}}}{3}, \qquad \text{where} \quad \mathcal{K}^2 \equiv c_K\, S^2. \tag{4}$$

The K^2 term from the UFE's left-hand side acts as a repulsive contribution on the right. For a torsional condensate sourced by polarized fermions, $S \propto a^{-3}$, leading to

$$H^2 = \frac{\kappa^2}{3}\Big[\rho_{\text{rad}}\, a^{-4} + \rho_{\text{m}}\, a^{-3}\Big] - \frac{c_K S_\star^2}{3}\, a^{-6} + \frac{\Lambda_{\text{eff}}}{3}. \tag{5}$$

As $a \to 0$, the repulsive a^{-6} term dominates and prevents collapse. The bounce ($H = 0$) occurs at a minimum scale factor

$$a_{\text{min}} = \begin{cases} \left(\frac{c_K S_\star^2}{\kappa^2 \rho_{\text{rad}}}\right)^{1/2} & \text{in the radiation era,} \\ \frac{c_K S_\star^2}{\kappa^2 \rho_{\text{m}}} & \text{in the matter era.} \end{cases} \tag{6}$$

The result is a non-singular bounce at a finite density, replacing the Big Bang singularity.

3 Black Holes as Spacetime Vortices and the Information Paradox

Regular Black Hole Solution. For a static, spherically symmetric metric with a stationary, vortex-like torsion profile, the UFE yields a Reissner–Nordström-like solution:

$$f(r) = 1 - \frac{2GM}{r} + \frac{Q_T^2}{r^2} - \frac{\Lambda_{\text{eff}}}{3} r^2. \tag{7}$$

Here, Q_T is an effective torsional charge arising from the K^2 term, which acts repulsively at short distances. For $Q_T \neq 0$, the central singularity is replaced by a regular core.

Information Storage. The torsional charge Q_T is topologically quantized and characterized by c_3. Information can be stored in this vortex topology. During evaporation, an extremal configuration ($M \to M_{\text{rem}} = \sqrt{Q_T^2/G}$) leads to a stable Planck-mass remnant. This topological memory provides a mechanism for resolving the information paradox within TFPTwithout violating unitarity.

4 Dark Matter: Relic Density of the TFPT Axion

Fixed Parameters from the E8 Cascade. The $n = 10$ block of the E8 cascade fixes the axion decay constant $f_a \simeq 8.86 \times 10^{10}\,\text{GeV}$ and mass:

$$m_a \simeq 64.36\,\mu\text{eV} \quad \Rightarrow \quad \nu_a = \frac{m_a}{2\pi\hbar} \approx 15.56\,\text{GHz}. \tag{8}$$

Geometrically Fixed Misalignment Angle. In TFPT, the cosmic birefringence $\beta = 2c_3\Delta a = \phi_0/(4\pi)$ implies a fixed initial displacement $\Delta a = \phi_0$. Identifying this with the QCD axion's misalignment angle θ_i leaves no freedom:

$$\boxed{\theta_i \quad = \quad \phi_0 \quad \approx \quad 0.05317}$$

Relic Density. With a small initial angle $\theta_i = \phi_0$, the standard pre-inflationary misalignment mechanism underproduces dark matter ($\Omega_a \ll \Omega_{\text{DM}}$). This is a sharp TFPT prediction: if the observed birefringence is sourced by a single global axion field, this axion can only be a sub-component of dark matter. The full dark matter density would require another source (e.g., torsional excitations) or an additional cold component.

Experimental Signature. A sharp spectral line target for haloscope experiments at $\nu \approx 15.56\,\text{GHz}$ with a fixed coupling $g_{a\gamma\gamma} = -4c_3$.

5 Dark Energy: Two TFPT-Consistent Paths

A) Torsional Condensate from the UFE. In the vacuum, the UFE $\left(R - \nabla\cdot K + K^2\right)_{AB} = 0$ for a homogeneous background with $\nabla\cdot K = 0$ implies a de Sitter solution. The effective cosmological constant is identified with the expectation value of the torsional term:

$$\boxed{\Lambda_{\text{eff}} \quad = \quad \tfrac{1}{4}\left\langle K^2 \right\rangle}$$

Thus, dark energy is not a new substance but the macroscopic manifestation of the universe's microscopic spacetime vorticity.

B) Terminal Stage of the E8 Cascade. The E8 cascade provides a discrete ladder of energy scales, $X_n \propto M_{\text{Pl}}\phi_n$. Assigning the vacuum energy density to a terminal stage of the cascade ($n_\star \geq 26$) yields

$$\rho_\Lambda \sim (M_{\text{Pl}}\phi_{n_\star})^4, \tag{9}$$

where the extreme suppression of $\phi_{n_\star}$ naturally explains the smallness of the observed value. Both paths are compatible: A) provides the geometric origin of Λ, while B) explains its magnitude.

6 Summary and Outlook

Key Results and Predictions. The TFPT framework is highly constrained. The cubic condition, the backreaction mechanism $\phi_0(\alpha)$, the UFE, and the birefringence formula are all fixed without free parameters.

- **Cosmic Birefringence:** $\beta = 0.2427^\circ$ is consistent with Planck PR4 data at the 0.5 to $1.7\,\sigma$ level.
- **Axion Window:** $m_a \simeq 64.36\,\mu\text{eV}$ ($\nu \approx 15.56\,\text{GHz}$) provides a sharp target. The misalignment angle $\theta_i = \phi_0$ is a prediction, not a free parameter.

New, Precise Tests.

1. Measurement of cosmic birefringence (EB and TB modes) with a fixed value and sign.
2. A haloscope search in the narrow band around 15.56 GHz.
3. Signatures of a bouncing cosmology from the a^{-6} torsion term in primordial spectra.
4. Astrophysical signatures of stable black hole remnants due to torsional charge.

References

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